



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
30-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

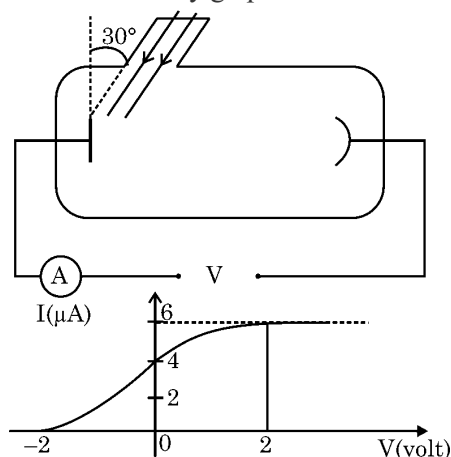
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In given experiment on the photoelectric effect monochromatic plane wave of light falls on a metal plate of area 66.3 cm^2 at angle 30° as shown. Electric field in light wave varies according to $E = E_0 \cos(\omega t + \phi_0)$ where amplitude $E_0 = 15 \text{ V/m}$, angular frequency is $\omega = 9.5 \times 10^{15} \text{ rad/s}$. Dependence of photoelectric current I on voltage applied V between the electrodes is shown by graph. Select the correct statement (s).

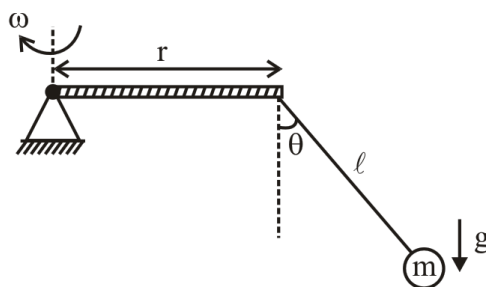


- (A) Work function of metal plate is 4.26 eV.
- (B) Power received by metal plate by radiation is $9.9 \times 10^{-4} \text{ W}$.
- (C) Quantum yield (photoelectric efficiency) in the experiment is 0.038.
- (D) If amplitude of electric field in incident light is made 25 V/m then saturation current will become 16.67 μA .

2. Select correct statement(s) :-

- (A) If two particle having same mass have same De-Broglie wavelength then they have same kinetic energy and same momentum.
- (B) When a bomb at rest explodes into two unequal pieces then both pieces have same De-Broglie wavelength.
- (C) When a photon of wavelength λ incident on a free electron at rest then it's De-Broglie wavelength becomes $\sqrt{\frac{h\lambda}{2mC}}$, where m is mass of electron.
- (D) A proton and an alpha particle are moving along circular path under same uniform magnetic field, then ratio of radius will be 2 : 1 if they have same De-Broglie wavelength

3. A rod of length $r = 3$ cm (exactly known) is pivoted about it's end and free to rotate in horizontal plane. A string of length $\ell = (20 \pm 1)$ cm is connected to opposite end of rod. A stone of mass m is connected to other end of string. When rod rotate at rate $\omega = (7 \pm 0.07)$ rad/s then string makes angle $\theta = \left(37 \pm \frac{3}{\pi}\right)^\circ$ with vertical. Then :

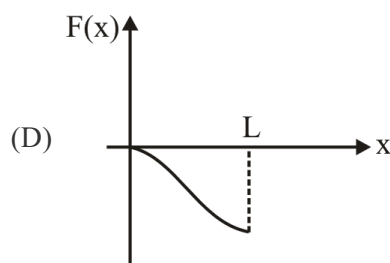
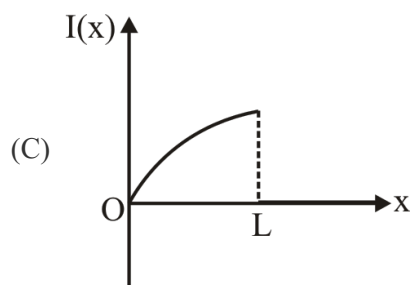
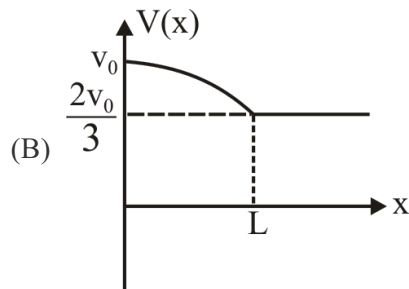
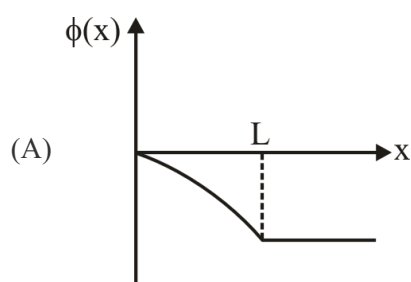
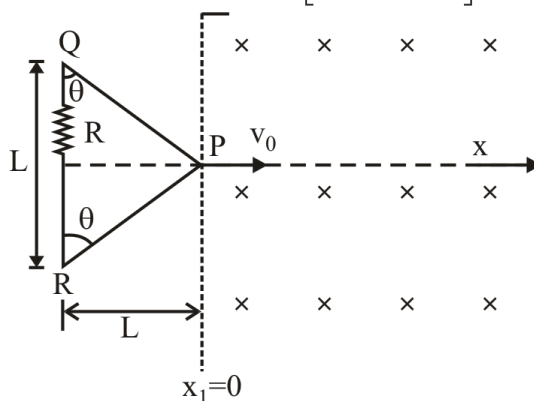


- (A) mean value of acceleration due to gravity is 9.8 m/s^2
- (B) mean value of acceleration due to gravity is 12.25 m/s^2
- (C) % error in measurement of g is nearly 7.67%
- (D) % error in measurement of g is nearly 5%

Space for Rough Work

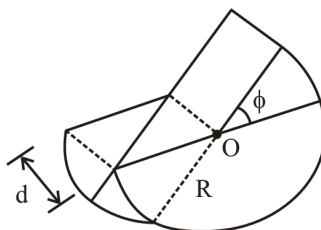
4. A rigid wire loop of equilateral triangle shape having resistance R is moving with constant velocity v_0 in plane of paper along x -axis. At $t = 0$ edge P of loop enter a region where there is a uniform magnetic field B in to plane of paper. For sufficiently large B , the loop eventually enters the region. Let x be the location of edge P . Let $\phi(x)$, $v(x)$, $I(x)$ and $F(x)$ represent magnetic flux through loop, velocity of loop, current in loop and force on the loop respectively as a function of x counter-clockwise direction is taken as positive.

Which of following schematic plot(s) is (are) correct ? $\left[\frac{B^2 L^3}{mR} = v_0 \right]$



Space for Rough Work

5. A capacitor consists of two stationary plates shaped as a semi-circle of radius R and a movable plate made of dielectric with permittivity ϵ and capable of rotating about an axis O between the stationary plates. The thickness of movable plate is d , which is nearly same as separation between stationary plates. Select correct statement(s). ϕ is angle of sector outside of capacitor.



- (A) Capacitance of system is proportional to $\left\{ \epsilon - \frac{\phi}{\pi} (\epsilon - \epsilon_0) \right\}$
- (B) If potential difference V is applied to capacitor electric torque about axis O is $\frac{V^2 R^2}{4d} (\epsilon - \epsilon_0)$ acting on movable plate.
- (C) If this capacitor is charged by charge Q and then source is removed then electric torque about axis O is $\frac{Q^2}{4(\epsilon - \epsilon_0)d}$.
- (D) If potential difference V is applied to capacitor and system is released than movable plate will perform simple harmonic motion.
6. A thin uniform disk of mass m and radius R suspended by an elastic thread in horizontal plane in a liquid. The torsional constant of thread is C and the resistance force acting per unit area of disk is $F = \eta v$ where η is a constant and v is the velocity of given element of disk. Take : $m = 1$ kg, $R = 0.5$ m, $C = 12.5$ Nm/rad.
- (A) Natural angular frequency of torsional oscillation is 10 rad/s
- (B) For $\eta \geq \frac{40}{\pi} \frac{\text{Ns}}{\text{m}^3}$, system will not be oscillatory.
- (C) For $\eta = \frac{24}{\pi} \frac{\text{Ns}}{\text{m}^3}$, system will perform damped oscillations with angular frequency 6 rad/s
- (D) For $\eta = \frac{24}{\pi} \frac{\text{Ns}}{\text{m}^3}$, Q-factor of system is $\frac{5}{6}$.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match list-I of nuclear process with list-II containing parent nucleus and one of end product of each process and then select the correct answer using codes given below the lists.

List-I		List-II	
(I)	Alpha decay	(P)	${}_{91}\text{Pa}^{230} \rightarrow {}_{90}\text{Th}^{230} + \dots\dots\dots$
(II)	β^{-} decay	(Q)	${}_{30}\text{Zn}^{57} \rightarrow {}_{29}\text{Cu}^{56} + \dots\dots\dots$
(III)	β^{+} decay	(R)	${}_{1}\text{H}^3 \rightarrow {}_{2}\text{He}^3 + \dots\dots\dots$
(IV)	neutron emission	(S)	${}_{53}\text{I}^{137} \rightarrow {}_{53}\text{I}^{136} + \dots\dots\dots$
		(T)	${}_{94}\text{Pu}^{239} \rightarrow {}_{92}\text{U}^{235} + \dots\dots\dots$

(A) I $\rightarrow$ T; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ R

(B) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ S

(C) I $\rightarrow$ T; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ S

(D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P

Space for Rough Work

8. List-I contain some optical arrangement and list-II contains deviation produced after all possible optical event in degree then select correct answer using following codes.

List-I		List-II	
(I)		(P)	0
(II)		(Q)	60°
(III)		(R)	1.8°
(IV)		(S)	178.2°
		(T)	179.4°

(A) I → R; II → P; III → Q; IV → R

(B) I → R; II → Q; III → Q; IV → S

(C) I → Q; II → P; III → S; IV → R

(D) I → R; II → Q; III → Q; IV → T

Space for Rough Work

9. Here I and V represent current and voltage respectively (steady state for DC and rms for AC) match list-I with list-II using given codes.

List-I		List-II	
(I)		(P)	$I = 0.2 \text{ A}$
(II)		(Q)	$I = 0 \text{ A}$
(III)		(R)	$V = 20 \text{ Volt}$
(IV)		(S)	$V = 0 \text{ Volt}$
		(T)	$V = 10 \text{ Volt}$

(A) $I \rightarrow \text{P,T}; \text{II} \rightarrow \text{P,R}; \text{III} \rightarrow \text{Q,R}; \text{IV} \rightarrow \text{P,S}$

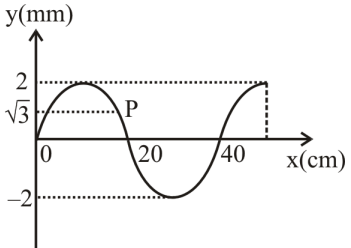
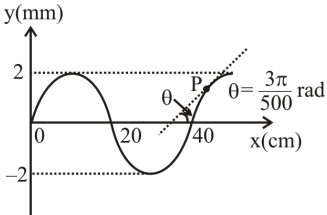
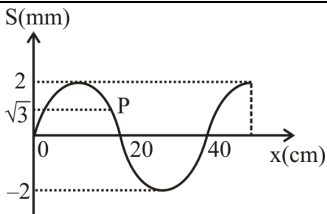
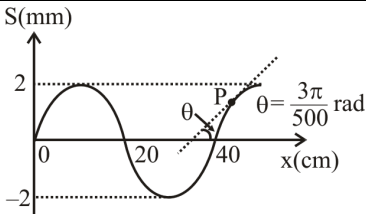
(B) $I \rightarrow \text{P,T}; \text{II} \rightarrow \text{P,R}; \text{III} \rightarrow \text{P,S}; \text{IV} \rightarrow \text{P,T}$

(C) $I \rightarrow \text{P,R}; \text{II} \rightarrow \text{P,S}; \text{III} \rightarrow \text{Q,R}; \text{IV} \rightarrow \text{Q,R}$

(D) $I \rightarrow \text{P,T}; \text{II} \rightarrow \text{P,R}; \text{III} \rightarrow \text{P,S}; \text{IV} \rightarrow \text{P,S}$

Space for Rough Work

10. List-I contain information about mechanical waves and list-II contain information about velocity of particle of medium corresponding to point P shown. Match list-I with list-II using given codes.

List-I		List-II	
(I)	 Transverse travelling wave moving in positive x-direction with speed 20 m/s	(P)	$v_p = \frac{3\pi}{50} \text{ m/s}$
(II)	 A transverse standing wave oscillating with frequency 50 Hz, having maximum amplitude 2.5 mm	(Q)	$v_p = \frac{\pi}{10} \text{ m/s}$
(III)	 A longitudinal wave propagating in negative x-direction with speed 20 m/s	(R)	moving in positive x-direction
(IV)	 A longitudinal standing wave oscillating with frequency 50 Hz, having maximum amplitude 2.5 mm.	(S)	moving in positive y-direction
		(T)	Insufficient information to find direction of motion

(A) I → Q,S; II → P,T; III → Q; IV → P,T

(B) I → Q,S; II → P; III → Q,R; IV → P,R

(C) I → Q,S; II → P,T; III → Q,T; IV → P

(D) I → P,T; II → Q,S; III → Q; IV → Q,T

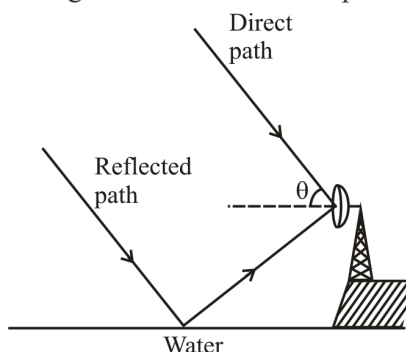
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

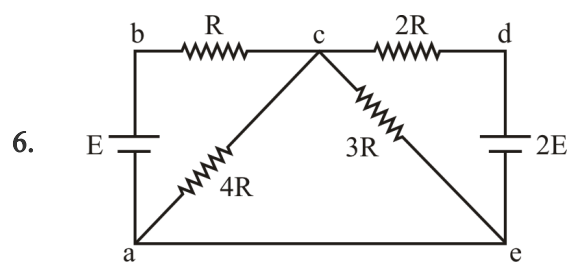
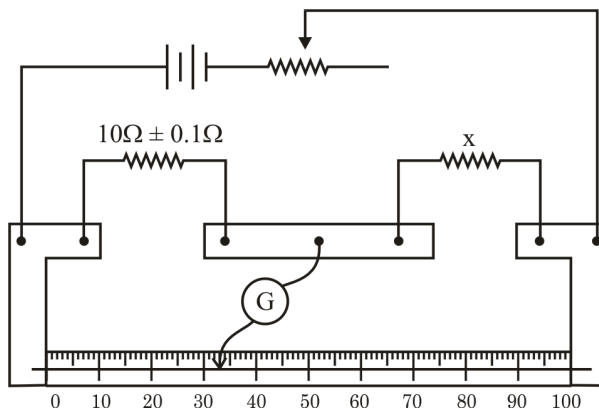
Zero Marks : 0 In all other cases.

1. Parallel beam of intensity I incident on a sphere, which absorb 50% of incident light and reflect remaining light. If radius of sphere is R then radiation force on sphere is given by $\frac{P}{Q} \frac{I (\pi R^2)}{c}$ then $P + Q$ is (where P and Q are smallest possible natural number) :-
2. A positron has 0.04 MeV kinetic energy. When it collide with an free electron at rest then they annihilate each other. One of the two photon produced is found to be moving in direction of incident positron. It's energy x MeV. Find x . Neglect any relativistic effect. [mass of electron = mass of positron = $\left(\frac{0.5}{C^2}\right)$ MeV]
3. Radio waves of wavelength 120m from a galaxy reach a radio telescope by two separate path as shown. As galaxy rises in east over water, the minimum intensity of destructive interference occurs when galaxy is $\theta = 37^\circ$ above the horizon. Find the height of the radio telescope disk above the water (in km).



Space for Rough Work

4. Two metal balls of same radius a are located in a homogenous poorly conducting medium with resistivity ρ . The resistance of medium between the balls is $\frac{\rho}{Na}$, assuming the separation between them is much greater than radius of the ball find N .
5. In meter bridge setup balanced length is measured by a meter scale having least count of 0.1 cm. When known resistance of $10\ \Omega \pm 0.1\ \Omega$ is using then balanced lengths are found to be 40.0 cm. Find % error in measurement of unknown resistance x .



6. Taking $R = 1\ \text{k}\Omega$ and $E = 250\ \text{V}$ find current in wire ae in mA .

7. A particle with charge q is located at $(-R, 0, 0)$ and a particle with charge $-2q$ is located at origin then equipotential surface that has zero potential occupy KR^3 volume in space. Find K .
8. Two stars are revolving in circular paths about center of mass of system. If their masses are M and $3M$ and separation between them is 4 times separation between the Earth and Sun. If the orbital period is 1 year find ratio of M and mass of Sun.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is/are coloured?

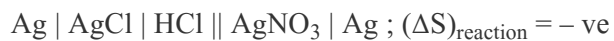
- (A) K_2MnO_4 (B) KO_2 (C) K_2CrO_4 (D) PbO_2

2. For phosphinic acid which option is/are correct?

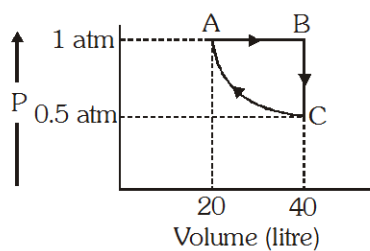
- (A) The ratio of total number of lone pair electrons & σ bond is 4/5
 (B) Two atoms are sp^3 hybridised
 (C) It is monobasic acid
 (D) Value of number of $p\pi-p\pi$ bond $\times$ value of number of $p\pi-d\pi$ bond = zero

Space for Rough Work

3. Select correct statement(s) for the cell with notation :

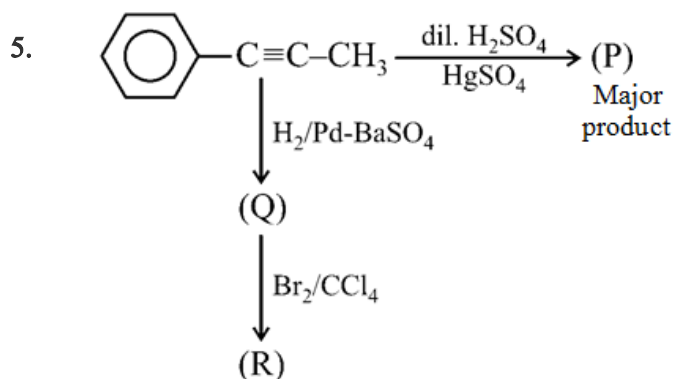


- (A) $E_{\text{cell}}^0 > 0$
 (B) $\Delta G^0 < 0$
 (C) E_{cell} decreases by increasing temperature
 (D) E_{cell} decreases by diluting anodic solution
4. Which option(s) is/are correct for given P – V plot for 2 moles of an ideal gas ?
 [Given CA is Isothermal process]



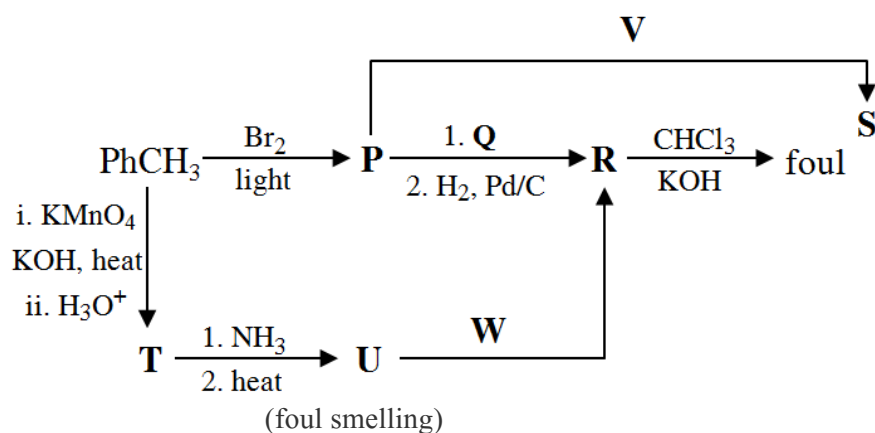
- (A) $W_{AB} = -20 \text{ L-atm}$
 (B) $\Delta U \text{ for Cycle} = 0$
 (C) $\Delta S_{\text{Cycle}} = 0$
 (D) $W_{CA} = +13.86 \text{ L atm}$

Space for Rough Work



Which of the following statement is/are correct for above reaction sequences.

- (A) Compound (P) can give aldol but do not give tollen's reagent test
 (B) (P) can give haloform test
 (C) (R) is a mixture of threo-vicinal dibromide
 (D) Ozonolysis of (Q) gives two products both of which give positive tollen's reagent test
6. Correct option(s) for the following sequence of reactions is(are)



- (A) $Q = KNO_2$, $W = LiAlH_4$ (B) $R = \text{benzenamine}$, $V = KCN$
 (C) $Q = AgNO_2$, $R = \text{phenylmethanamine}$ (D) $W = LiAlH_4$, $V = AgCN$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.

	List-I		List-II
(P)	$\text{H}_4\text{P}_2\text{O}_7$	(1)	$\text{p}\pi\text{-d}\pi$ bond is present
(Q)	P_4O_7	(2)	P–O–P is present
(R)	P_4O_6	(3)	'P' is sp^3 hybridised
(S)	P_4O_8	(4)	Each 'P' has 4 σ & 1 π bond
		(5)	Each 'P' has only 3 σ bond

(A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 1; \text{R} \rightarrow 5; \text{S} \rightarrow 3$

(B) $\text{P} \rightarrow 4; \text{Q} \rightarrow 5; \text{R} \rightarrow 5; \text{S} \rightarrow 4$

(C) $\text{P} \rightarrow 5; \text{Q} \rightarrow 5; \text{R} \rightarrow 4; \text{S} \rightarrow 4$

(D) $\text{P} \rightarrow 5; \text{Q} \rightarrow 5; \text{R} \rightarrow 4; \text{S} \rightarrow 3$

8.

	List-I		List-II
(P)	PbI_2	(1)	Water insoluble compound
(Q)	AgI	(2)	Yellow colour
(R)	Hg_2I_2	(3)	Green colour
(S)	HgI_2	(4)	Scarlet red colour
		(5)	White colour

(A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 4$

(B) $\text{P} \rightarrow 4; \text{Q} \rightarrow 3; \text{R} \rightarrow 2; \text{S} \rightarrow 2$

(C) $\text{P} \rightarrow 5; \text{Q} \rightarrow 3; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

(D) $\text{P} \rightarrow 5; \text{Q} \rightarrow 3; \text{R} \rightarrow 2; \text{S} \rightarrow 1$

Space for Rough Work

9. Answer the following by appropriately matching the lists based on the information given in the list.

	List-I		List-II
(P)	$\text{Zn} \text{Zn}^{2+}(\text{C}) \text{Zn}^{2+}(\text{2C}) \text{Zn}$	(1)	Spontaneous cell reaction
(Q)	$\text{H}_2(\text{P}) \text{HCl}(1\text{N}) \text{H}_2\text{SO}_4(1\text{N}) \text{H}_2(\text{P})$	(2)	According to cell representation electric work is produced.
(R)	$\text{Cu} \text{Cu}^{2+}(0.01\text{M}) \text{Ag}^+(0.1\text{M}) \text{Ag}$	(3)	Concentration of cation in cathodic compartment increases upto equilibrium
(S)	$\text{Ag} \text{AgCl}(\text{KCl}, 0.1\text{M}) \text{Ag}^+(0.01\text{M}) \text{Ag}$	(4)	Concentration cell
		(5)	Concentration of Cl^- in anodic compartment decreases

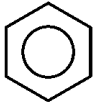
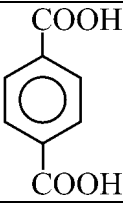
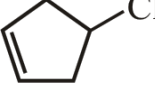
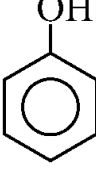
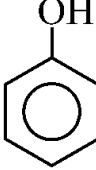
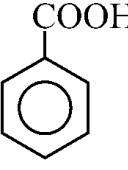
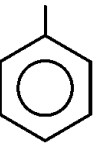
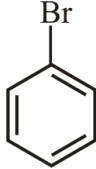
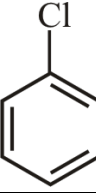
Correctly match among the following is :

[Given : $E_{\text{Ag}^+/\text{Ag}}^0 = 0.8\text{ V}$, $E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34\text{ V}$, $K_{\text{sp}}(\text{AgCl}) = 10^{-10}$]

- (A) $\text{P} \rightarrow 1,3; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 1,2; \text{S} \rightarrow 1,2,3,4$
 (B) $\text{P} \rightarrow 1,2,4; \text{Q} \rightarrow 3,5; \text{R} \rightarrow 1,3; \text{S} \rightarrow 1,2,4$
 (C) $\text{P} \rightarrow 1,2,4; \text{Q} \rightarrow 4; \text{R} \rightarrow 1,2; \text{S} \rightarrow 1,2,5$
 (D) $\text{P} \rightarrow 1,2,3; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 1,2,5; \text{S} \rightarrow 1,2,3,5$

Space for Rough Work

10.

List-I		List-II	
(P)	 $\xrightarrow[\text{(iii) H}^+]{\text{(i) } \text{CH}_3\text{CHClCH}_3 / \text{Anhy. AlCl}_3, \text{(ii) O}_2 / h\nu}$	(1)	
(Q)	 $\xrightarrow[\text{(iii) CHCl}_2\text{Br / KOH}]{\text{(i) aq. KOH, (ii) H}^+ / \Delta}$	(2)	
(R)	 $\xrightarrow[\text{(ii) Zn-dust}]{\text{(i) CCl}_4 / \text{NaOH}}$	(3)	
(S)	 $\xrightarrow[\text{(ii) KMnO}_4]{\text{(i) CH}_3\text{COCl / Anhy. AlCl}_3}$	(4)	
		(5)	

The correct option is :

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

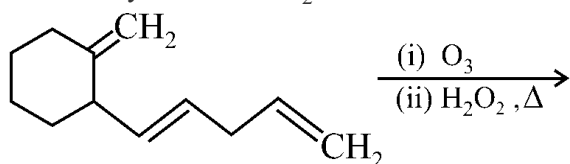
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

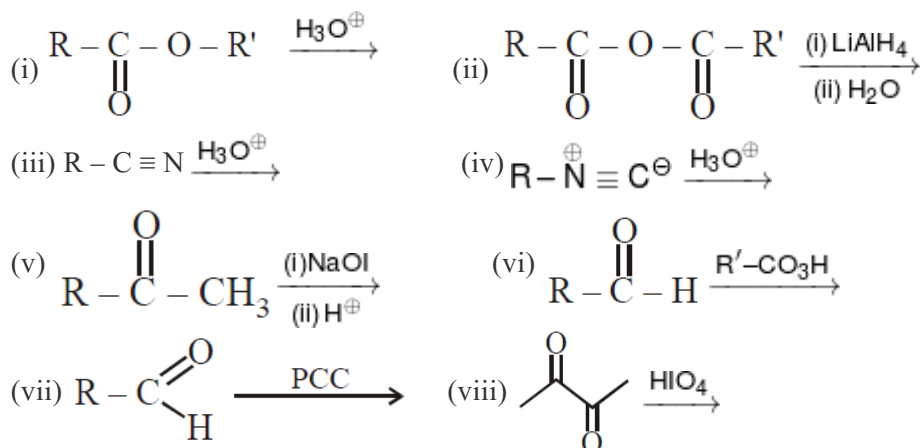
1. For $K_2Cr_2O_7$,
 Total number of σ -bond = x
 Total number of π -bond = y
 Total number of lone pair electron = a
 Find the value of (x + y + a)
2. For Borax Crystalline, $(Na_2B_4O_7 \cdot 10H_2O)$
 Total number of σ bond = a
 Total number of sp^3 hybridised atom = b
 Total number of lone pair on non metal atoms = c
 Total number of sp^2 hybridised atom = d
 $\frac{a}{c} \times \frac{b}{d} = z$
 Find the value of 'z'.
3. The molar conductivity of a solution of a weak acid HX (0.1 M) is 100 times smaller than the molar conductivity of a solution of a weak acid HY (0.1 M). If $\lambda_{X-}^0 = \lambda_{Y-}^0$, the difference in their pK_a values, $pK_a(HX) - pK_a(HY)$, is (consider degree of ionization of both acids to be $\ll 1$).
4. The time for half-life period of a certain reaction. $A \rightarrow$ products is 1 h, when the initial concentration of the reactant 'A', is 2.0 mol L^{-1} . How much time does it take (in hours) for its concentration to come from 0.50 to 0.25 mol L^{-1} , if it is a zero order reaction?

Space for Rough Work

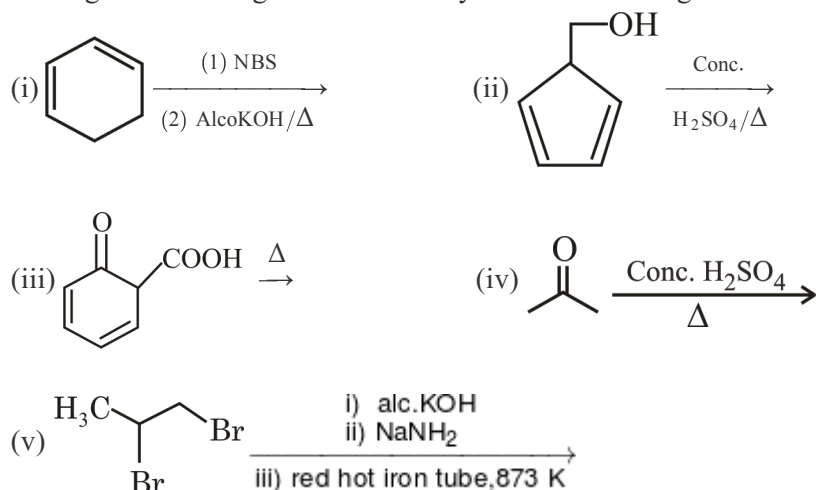
5. 0.1 M aqueous solution of a weak base BOH has pH = 9.5. Find pK_b of the weak base, BOH ?
6. How many moles of CO_2 are released in following reaction :



7. Total number of reactions in which carboxylic acid is obtained as a final product.



8. Among the following reaction Identify reactions which give Aromatic product ?



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let A, B, C be the three sets of complex numbers as defined below $A = \{Z : \text{Im } Z \geq 1\}$, $B = \{Z : |Z - 2 - i| = 3\}$ and $C = \{Z : \text{Re}((1 - i)Z) = \sqrt{2}\}$. Let 'Z' be any point in $A \cap B \cap C$ and let ' ω ' be any point satisfying $|\omega - 2 - i| < 3$, then which of the following may be correct ?

(A) $|Z| < 3$

(B) $|\omega| < 4$

(C) $|Z| - |\omega| > -6$

(D) $|Z| > 3$

Space for Rough Work

2. Consider the planes $P_1 : 2x + y + z + 4 = 0$, $P_2 : y - z + 4 = 0$ and $P_3 : 3x + 2y + z + 8 = 0$. Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3 , P_3 and P_1 & P_1 and P_2 respectively. Then,
- (A) Atleast two of the lines L_1, L_2 and L_3 are non-parallel
 (B) Atleast two of the lines L_1, L_2 and L_3 are parallel
 (C) The three planes intersect in a line
 (D) The three planes form a triangular prism
3. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$, then
- (A) $\lim_{x \rightarrow 0} f(x) = 1$
 (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$
 (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity
 (D) $f(x)$ is an even function
4. Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as
 E_1 : Their product is divisible by 5
 E_2 : Unit's place in their product is 5.
 Which of the following statement(s) is/are correct ?
- (A) E_1 is twice as likely to occur as E_2
 (B) E_1 and E_2 are disjoint
 (C) $P(E_2/E_1) = 1/4$
 (D) $P(E_1/E_2) = 1$

Space for Rough Work

5. Let $f(x) = 2x^3 - 15x^2 + 36x - 23$ & $g(x) = \begin{cases} \max \{f(t), 1 \leq t \leq x\} & , \text{ if } 1 \leq x \leq \frac{7}{2} \\ 12 - 2x & , \text{ if } \frac{7}{2} < x \leq 6 \end{cases}$. Which of the following is correct about $g(x)$.
- (A) Continuous everywhere
 (B) Differentiable everywhere
 (C) has greatest value 5
 (D) Not differentiable at 2 points
6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(|f(x) - f(y)|) = f(f(x)) - 2x^2f(y) + f(y^2)$ where $f(x)$ is not identically zero everywhere. $f(x)$ can not be
- (A) ONE - ONE
 (B) ONTO
 (C) Monotonic
 (D) MANY-ONE

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7.	List-I	List-II
(I)	$A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ is inverse of $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ 1 & -4x & -2x \end{bmatrix}$, then the value of x is ?	(P) 3
(II)	If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $ A^3 = 125$ then α is equal to ($\alpha > 0$)	(Q) $\frac{1}{5}$
(III)	If a, b, c are three complex numbers such that $a^2 + b^2 + c^2 = 0$ and $\Delta = \begin{vmatrix} b^2 + c^2 & ab & ac \\ ab & c^2 + a^2 & bc \\ ac & bc & a^2 + b^2 \end{vmatrix} = ka^2b^2c^2$	(R) 1
(IV)	If $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$ then $f\left(\frac{2\pi}{3}\right)$ is	(S) 4
		(T) 2

(A) I → R; II → R; III → T; IV → T

(B) I → R; II → P; III → T; IV → R

(C) I → Q; II → P; III → S; IV → R

(D) I → Q; II → P; III → S; IV → T

Space for Rough Work

8.

List-I		List-II	
(I)	If the line $\frac{x-1}{\beta} = \frac{y-\alpha}{-6} = \frac{z-1}{-2}$ entirely lies in the plane $2x + y - 6z = 0$, then $\alpha + \beta$ is equal to	(P)	-2
(II)	If planes $2x - y + z = 2$, $x - 3y + z = \beta$ & $\alpha x + y + z = 5$ have no common point, then β cannot be	(Q)	1
(III)	If a, b, c are +ve real numbers and $ax + by + z = 0$, $x + by + cz = 0$ & $ax + y + cz = 0$ have infinite common points, then $2abc - ab - bc - ca$ is equal to	(R)	-1
(IV)	Equation of plane through (2, 1, 4) and perpendicular to the line $3x - y + z + 1 = 0 = 5x + y + 3z$ is $\alpha x + y + \beta z + \gamma = 0$ then $\alpha + \beta + \gamma$ is equal to	(S)	2
		(T)	4

(A) I → Q; II → R; III → R; IV → T

(B) I → R; II → Q; III → R; IV → S

(C) I → Q; II → R; III → S; IV → P

(D) I → R; II → Q; III → S; IV → P

Space for Rough Work

9. A showroom has 10 different pairs of expensive shoes. A thief steals 4 shoes randomly from the showroom. Match the following :

List-I		List-II	
(I)	The probability that there is no matching pair among the 4 shoes stolen is	(P)	$\frac{3}{19 \times 17}$
(II)	The probability that there is atleast one matching pair among the 4 shoes stolen is	(Q)	$\frac{96}{19 \times 17}$
(III)	The probability that there is exactly one matching pair among the 4 shoes stolen is	(R)	$\frac{{}^{20}C_4 - {}^{10}C_4 \cdot 2^4}{{}^{20}C_4}$
(IV)	The probability that there are two matching pair among the 4 shoes stolen is	(S)	$\frac{14 \times 16}{19 \times 17}$
		(T)	$\frac{16}{19 \times 17}$

(A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

(B) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ Q

(C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ Q; IV $\rightarrow$ P

(D) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ P

Space for Rough Work

10.

List-I		List-II	
(I)	$\int_0^{\frac{\pi}{4}} \ln(\sin^2 2x) dx$	(P)	2
(II)	$\int_0^{\frac{\pi}{2}} (x \sin 2x - 1) e^{\cos^2 x} dx$	(Q)	$-\frac{\pi}{2}$
(III)	If $f\left(x + \frac{\pi}{24}\right) + f\left(x + \frac{5\pi}{24}\right) = f\left(x + \frac{3\pi}{24}\right)$ and $\int_{\frac{\pi}{4}}^{\frac{7\pi}{4}} f(x) dx = 3$ then $\int_0^{\pi} f(x) dx$ is equal to	(R)	$-\frac{\pi}{2} \ln 2$
(IV)	If $f(x) = \int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx$ & $f(0) = 1$ then $f(-1) =$	(S)	$\frac{\pi}{2}$
		(T)	1

(A) I $\rightarrow$ Q; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ T

(B) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ P; IV $\rightarrow$ P

(C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ T; IV $\rightarrow$ P

(D) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The value of $\frac{4 \int_0^{\infty} x^{10} e^{-x^2} dx}{7 \int_0^{\infty} x^6 e^{-x^2} dx}$ is
2. The minimum area bounded by the differentiable function $y = f(x)$ and $y = ax + 9$ ($a \in \mathbb{R}$) where f satisfies the relation $f(x+y) = f(x) + f(y) + y\sqrt{f(x)}$, $x, y \in \mathbb{R}$ and $f'(0) = 0$ is
3. a, b, c are cube roots of p ($p < 0$) then for any permissible real value of x, y, z it is given that $\left| \frac{xa + yb + zc}{xb + yc + za} \right| + (a_1^2 - 2b_1^2)\omega + ([x] + [y] + [z])\omega^2 = 0$ where ω is the imaginary cube root of unity and a_1, b_1 are prime numbers. Then $[x + a_1] + [y + b_1] + [z]$ is equal to (where $[.]$ denotes G.I.F.)
4. $\left| \lim_{x \rightarrow 0} \left[\lim_{n \rightarrow \infty} \left\{ \frac{n}{1 - n^2 x^2} + \frac{n}{4 - n^2 x^2} + \dots + \frac{n}{n^2 - n^2 x^2} \right\} \right] \right|$ is equal to

Space for Rough Work

5. The position vectors of two points A and C are $9\hat{i} - \hat{j} + 7\hat{k}$ and $7\hat{i} - 2\hat{j} + 7\hat{k}$ respectively $\overrightarrow{AB} = 4\hat{i} - \hat{j} + 3\hat{k}$, $\overrightarrow{CD} = 2\hat{i} - \hat{j} + 2\hat{k}$, point of intersection of lines AB and CD is P. Vector $\overrightarrow{PQ}$ is perpendicular to $\overrightarrow{AB}$ and $\overrightarrow{CD}$ and $PQ = 15$. If position vector of Q is $6\hat{i} - 9\hat{j} - \mu\hat{k}$ then value of μ is
6. If $Z = \frac{1}{2}(\sqrt{3} - i)$ then the least positive integral value of 'n' such that $(Z^{101} + i^{109})^{106} = Z^n$ is (where $i^2 = -1$)
7. In a multiple choice question there are four alternative answers of which one or more than one is correct. A candidate will get marks on the question only if he ticks all the correct answer. The candidate ticks the answers at random. The least number of chances he should be given if the probability of getting marks is greater than or equal to $\frac{1}{3}$ is -
8. Find the distance of the point $\hat{i} + 2\hat{j} + 3\hat{k}$ from the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5$ measured parallel to the vector $2\hat{i} + 3\hat{j} - 6\hat{k}$.

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work